# STATISTICS AND PROBABILITY

ST331

(Statistical Computing 3)

# ELEMENTARY STATISTICS

- A numeric variable is one whose observations are naturally recorded as numbers. There are two types of numeric variables: continuous and discrete
- Categorical variables are those that provide groupings that may have no logical order, or a logical order with inconsistent differences between groups

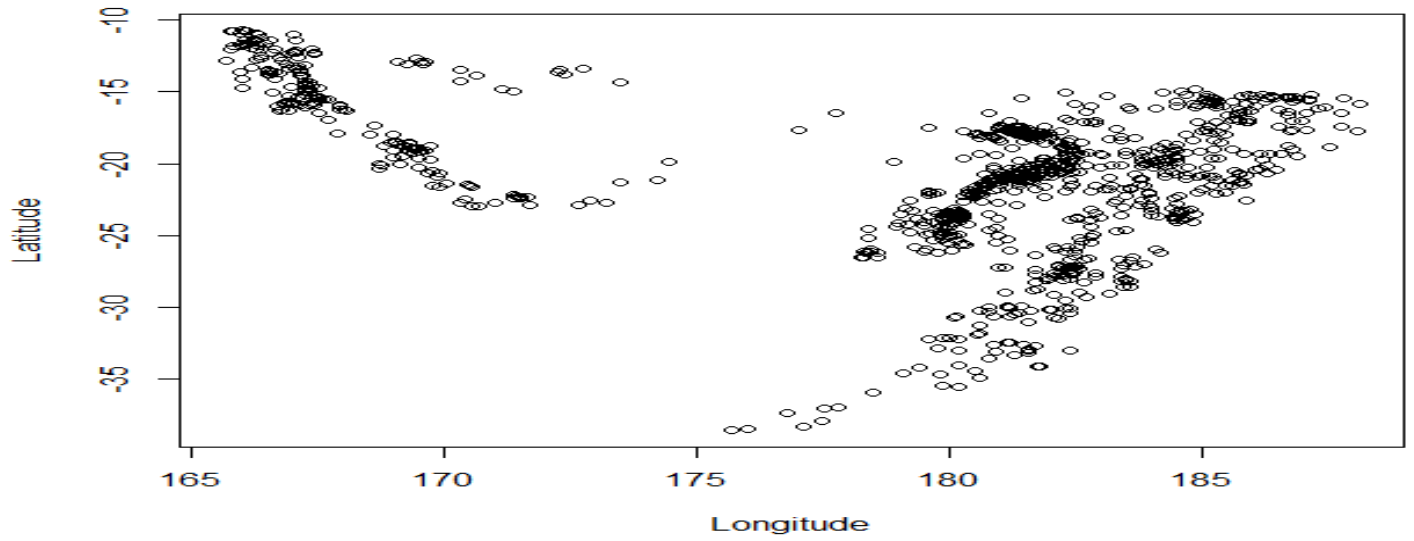
# ELEMENTARY STATISTICS

- weight is a is a numeric-continuous variable while feed is clearly a categorical variable because it has only six possible outcomes, which aren't numeric.

```
> chickwts[1:5,]
  weight      feed
1    179 horsebean
2    160 horsebean
3    136 horsebean
4    227 horsebean
5    217 horsebean
> chickwts$weight
 [1] 179 160 136 227 217 168 108 124 143 140 309 229 181 141 260 203 148
[18] 169 213 257 244 271 243 230 248 327 329 250 193 271 316 267 199 171
[35] 158 248 423 340 392 339 341 226 320 295 334 322 297 318 325 257 303
[52] 315 380 153 263 242 206 344 258 368 390 379 260 404 318 352 359 216
[69] 222 283 332
> chickwts$feed
 [1] horsebean horsebean horsebean horsebean horsebean horsebean
 [7] horsebean horsebean horsebean horsebean linseed   linseed
[13] linseed   linseed   linseed   linseed   linseed   linseed
[19] linseed   linseed   linseed   linseed   soybean   soybean
[25] soybean   soybean   soybean   soybean   soybean   soybean
[31] soybean   soybean   soybean   soybean   soybean   soybean
[37] sunflower sunflower sunflower sunflower sunflower sunflower
[43] sunflower sunflower sunflower sunflower sunflower sunflower
[49] meatmeal  meatmeal  meatmeal  meatmeal  meatmeal  meatmeal
[55] meatmeal  meatmeal  meatmeal  meatmeal  meatmeal  casein
[61] casein    casein    casein    casein    casein    casein
[67] casein    casein    casein    casein    casein
Levels: casein horsebean linseed meatmeal soybean sunflower
```

# ELEMENTARY STATISTICS

```
> quakes[1:5,]
     lat   long depth mag stations
1 -20.42 181.62   562 4.8       41
2 -20.62 181.03   650 4.2       15
3 -26.00 184.10    42 5.4       43
4 -17.97 181.66   626 4.1       19
5 -20.42 181.96   649 4.0       11
> plot(quakes$long,quakes$lat,xlab="Longitude",ylab="Latitude")
```



# CENTRALITY: MEAN, MEDIAN, MODE

```
> xdata <- c(2,4.4,3,3,2,2.2,2,4)
> x.bar <- mean(xdata)
> x.bar
[1] 2.825
> m.bar <- median(xdata)
> m.bar
[1] 2.6
> xtab <- table(xdata)
> xtab
xdata
  2 2.2   3   4 4.4
  3   1   2   1   1
> min(xdata)
[1] 2
> max(xdata)
[1] 4.4
> d.bar <- xtab[xtab==max(xtab)]
> d.bar
2
3
```

Here, 2 is the value and 3 is the frequency of that value.

# CENTRALITY: MEAN, MEDIAN, MODE

```
> mean(quakes$mag)
[1] 4.6204
> median(quakes$mag)
[1] 4.6
> Qtab <- table(quakes$mag)
> Qtab

  4 4.1 4.2 4.3 4.4 4.5 4.6 4.7 4.8 4.9   5 5.1 5.2 5.3 5.4 5.5 5.6 5.7 5.9   6 6.1 6.4
 46  55  90  85 101 107 101  98  65  54  47  43  29  21  20  14   9   8   2   3   1   1
> Qtab[Qtab==max(Qtab)]
4.5
107
```

# COUNTS, PERCENTAGES, AND PROPORTIONS

```
> table(chickwts$feed)

   casein horsebean   linseed  meatmeal   soybean sunflower
       12         10        12        11        14        12
> table(chickwts$feed)/nrow(chickwts)

   casein horsebean   linseed  meatmeal   soybean sunflower
0.1690141 0.1408451 0.1690141 0.1549296 0.1971831 0.1690141
> table(chickwts$feed)

   casein horsebean   linseed  meatmeal   soybean sunflower
       12         10        12        11        14        12
> table(chickwts$feed)/nrow(chickwts)*100

   casein horsebean   linseed  meatmeal   soybean sunflower
 16.90141  14.08451  16.90141  15.49296  19.71831  16.90141
> table(chickwts$feed)/nrow(chickwts)

   casein horsebean   linseed  meatmeal   soybean sunflower
0.1690141 0.1408451 0.1690141 0.1549296 0.1971831 0.1690141
> round(table(chickwts$feed)/nrow(chickwts),digits=3)

   casein horsebean   linseed  meatmeal   soybean sunflower
    0.169     0.141     0.169     0.155     0.197     0.169
```

# QUANTILES, PERCENTILES, AND THE FIVE NUMBER SUMMARY

```
> quantile(xdata,prob=0.8)
80%
3.6
> quantile(xdata,prob=c(0,0.25,0.5,0.75,1))
  0%  25%  50%  75% 100%
2.00 2.00 2.60 3.25 4.40
```

Here, you've used quantile to obtain what's called the five-number summary of xdata, comprised of the 0th percentile (the minimum), the 25th percentile, the 50$^{th}$ percentile, the 75th percentile, and the 100th percentile(the maximum).

```
> summary(xdata)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  2.000   2.000   2.600   2.825   3.250   4.400
> quantile(chickwts$weight,prob=c(0.25,0.75))
   25%    75%
204.5  323.5
```

This indicates that 25 percent of the weights lie at or below 204.5 grams and that 75 percent of the weights lie at or below 323.5 grams

# COVARIANCE AND CORRELATION

Correlation measures the extent (strength) and direction of the relationship, while covariance only shows the direction, not the extent

Covariance

$$r_{xy} = \frac{1}{n-1}\sum_{i-1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Correlation

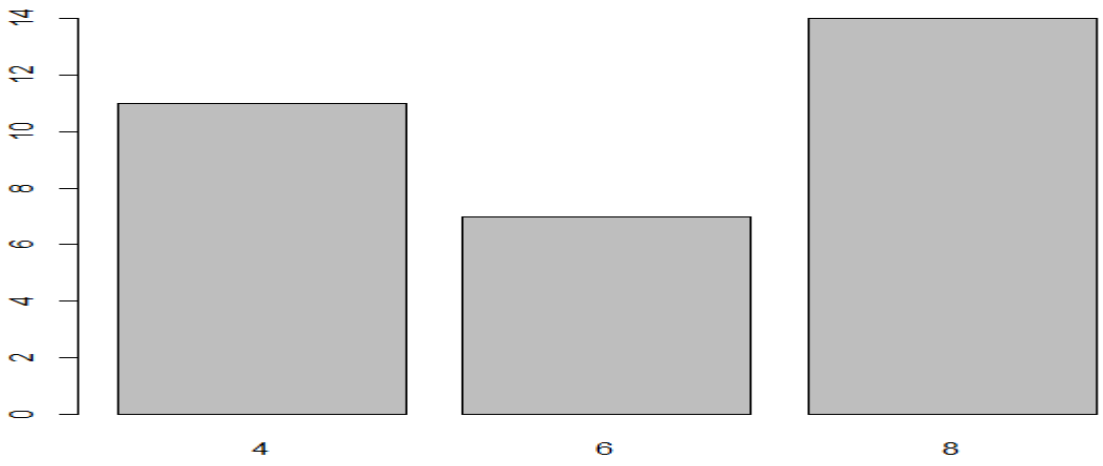
$$\rho_{xy} = \frac{r_{xy}}{s_x s_y},$$

```
> cov(xdata,ydata)
[1] 1.479286
> cor(xdata,ydata)
[1] 0.7713962
```

# BASIC DATA VISUALIZATION (BARPLOTS AND PIE CHARTS)

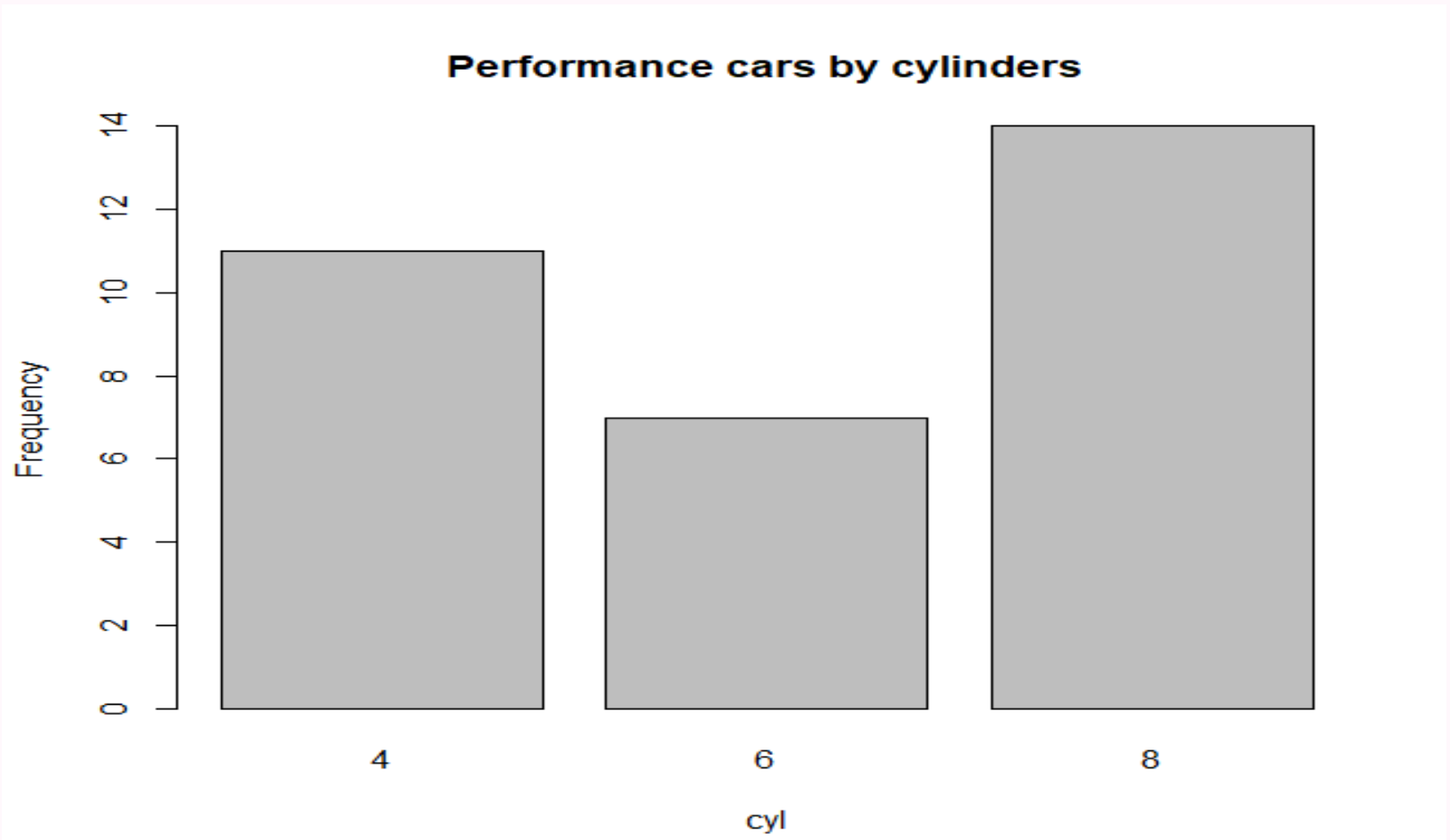
```
> mtcars[1:5,]
                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
> cyl.freq <- table(mtcars$cyl)
> cyl.freq

 4  6  8
11  7 14
> barplot(cyl.freq)
```



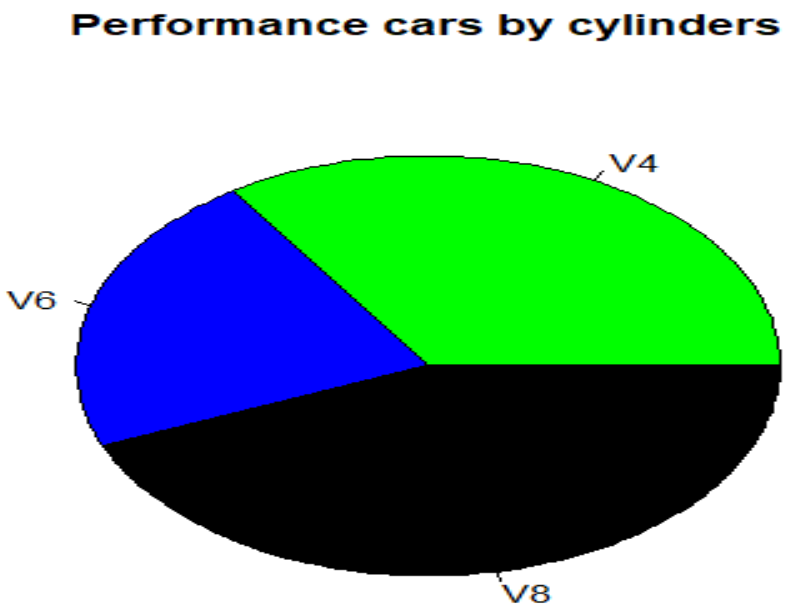
# BASIC DATA VISUALIZATION (BARPLOTS AND PIE CHARTS)

```
> barplot(cyl.freq, xlab="cyl", ylab="Frequency", main = "Performance cars by cylinders")
```



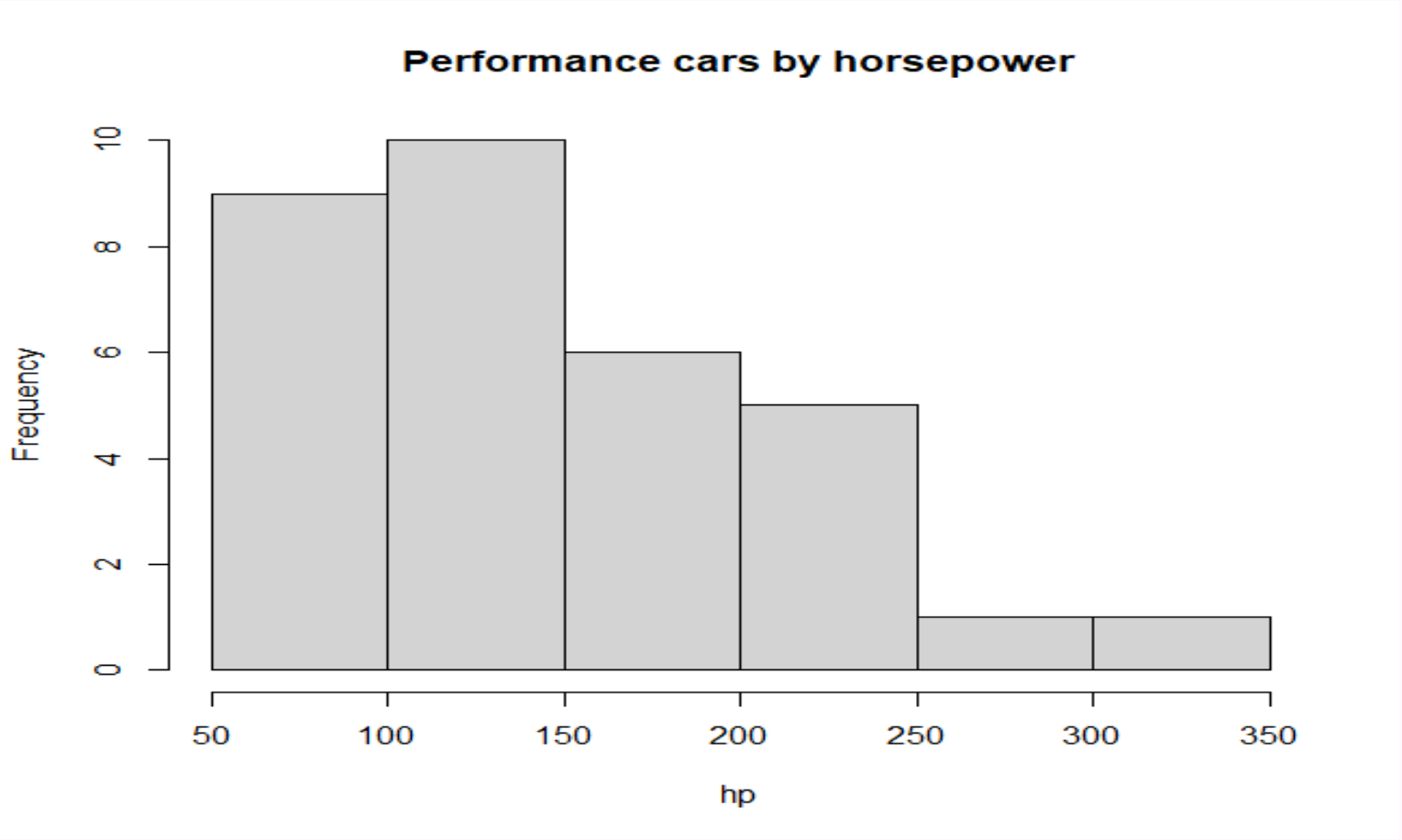
# BASIC DATA VISUALIZATION (BARPLOTS AND PIE CHARTS)

```
> pie(table(mtcars$cyl),labels=c("V4","V6","V8"),
+     col=c("green","blue","black"),main="Performance cars by cylinders")
```



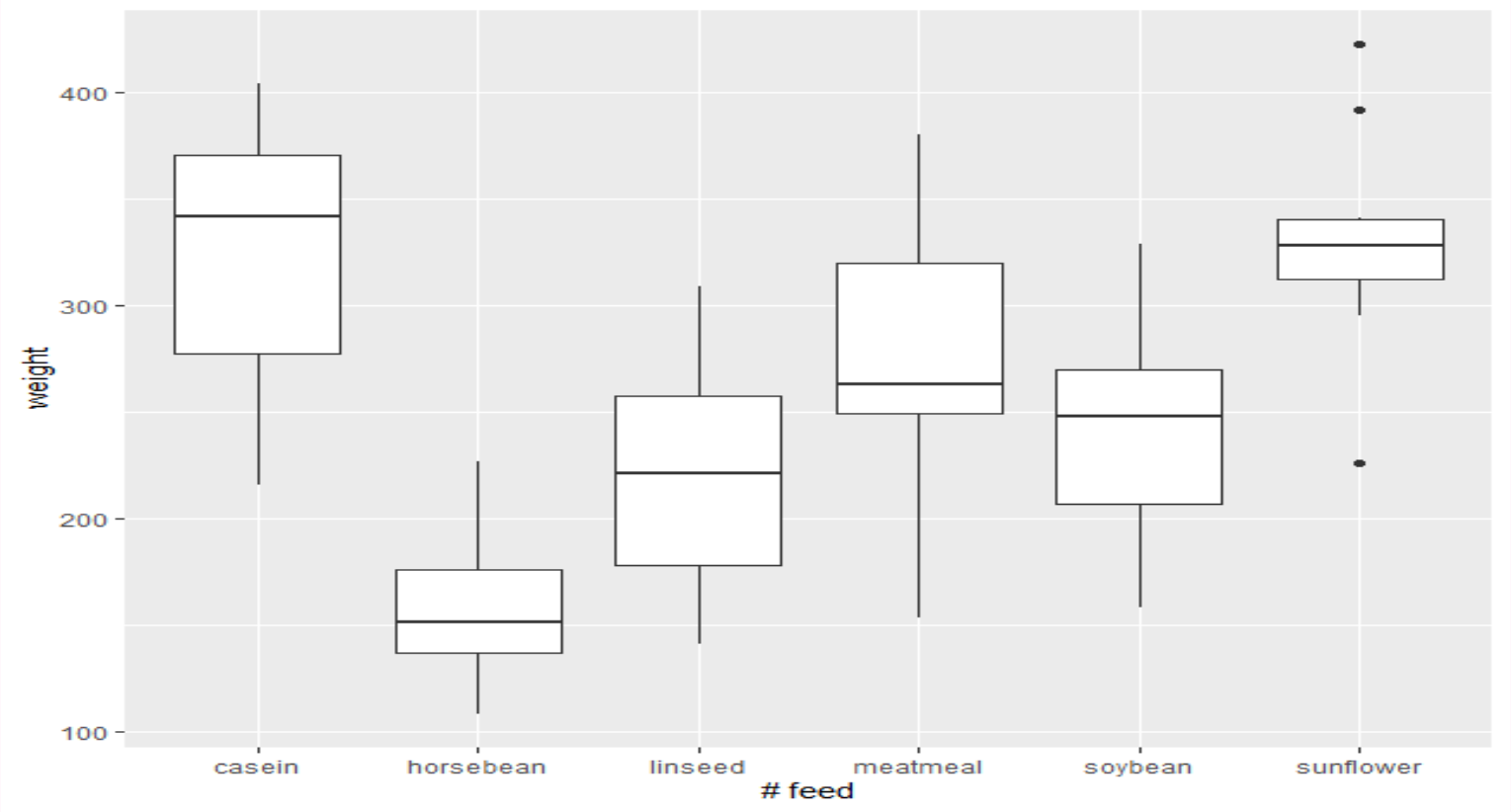
# HISTOGRAM

```
> hist(mtcars$hp, xlab="hp", ylab="Frequency", main = "Performance cars by horsepower")
```



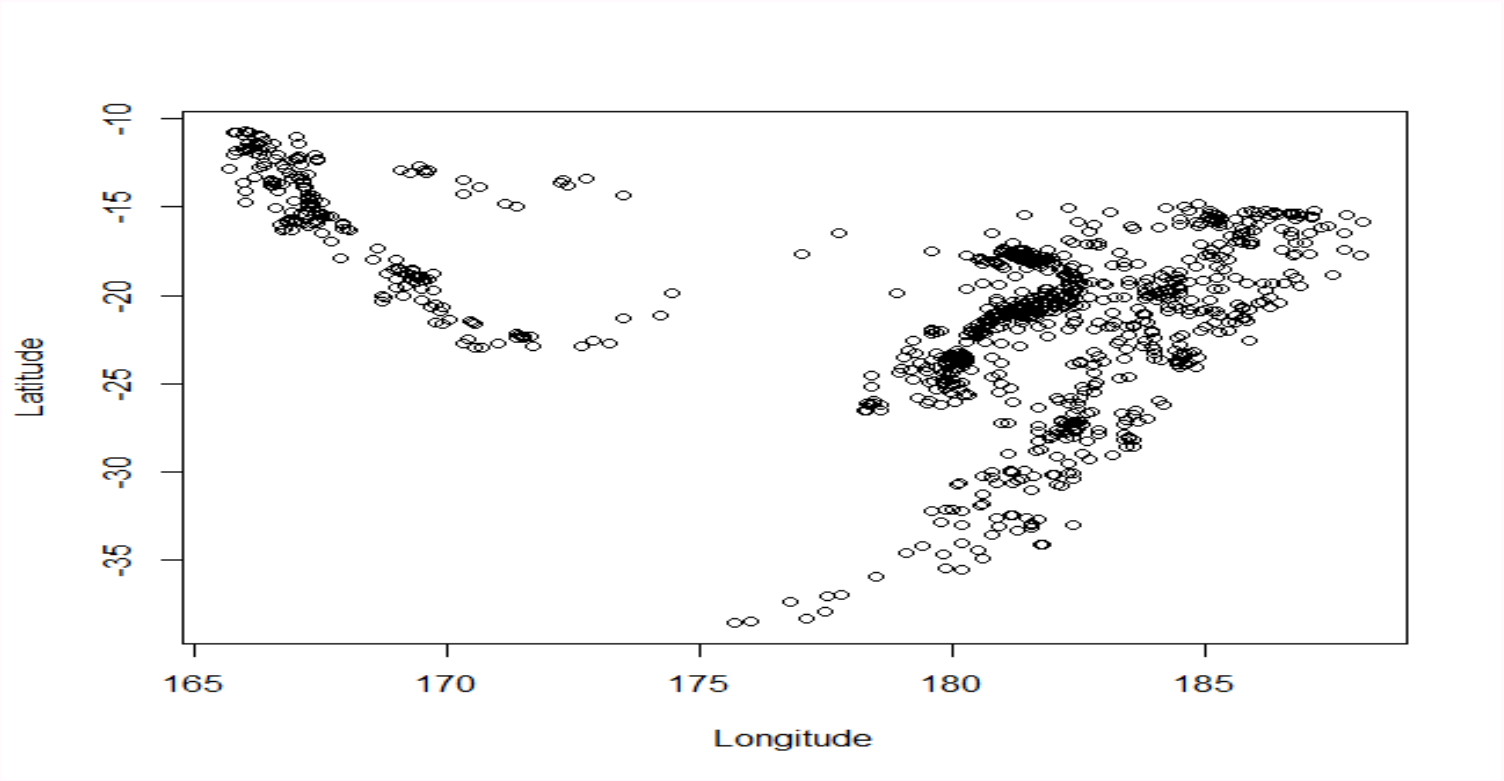
# BOXPLOT

```
> library(ggplot2)
> qplot(chickwts$feed,chickwts$weight,geom="boxplot",xlab="# feed",ylab="weight")
```



# SCATTERPLOTS

```
> plot(quakes$long,quakes$lat,xlab="Longitude",ylab="Latitude")
```



# PROBABILITY

**Intersection**

When a dice is toss, what is the probability that on a single toss you roll an even number and it's a 4 or more?

```
> P_ANB= (2/3)*(1/2)
> P_ANB
[1] 0.3333333
```

**Union**

So, in rolling the die, what's the probability that you observe an even number or one that's at least 4

```
> P_AUB= (1/2)+(1/2)-(1/3)
> P_AUB
[1] 0.6666667
```

**Complement**

find the probability that you do not roll a 4 or greater

```
> P_A_= 1-1/2
> P_A_
[1] 0.5
```

# CUMULATIVE PROBABILITY DISTRIBUTIONS OF DISCRETE RANDOM VARIABLES

**A cumulative probability for a random variable X is "the probability of observing less than or equal to x" and written as Pr(X ≤ x)**

| $x$ | −4 | 0 | 1 | 8 |
|---|---|---|---|---|
| $\Pr(X = x)$ | 0.32 | 0.48 | 0.15 | 0.05 |
| $\Pr(X \leq x)$ | 0.32 | 0.80 | 0.95 | 1.00 |

```
> X.outcomes <- c(-4,0,1,8)
> X.prob <- c(0.32,0.48,0.15,0.05)
> X.cumul <- cumsum(X.prob)
> X.cumul
[1] 0.32 0.80 0.95 1.00
```

# MEAN AND VARIANCE OF A DISCRETE RANDOM VARIABLE

## Mean

$$\mu_X = \mathbb{E}[X] = x_1 \times \Pr(X = x_1) + \ldots + x_k \times \Pr(X = x_k)$$
$$= \sum_{i=1}^{k} x_i \Pr(X = x_i)$$

```
> mu.X <- sum(X.outcomes*X.prob)
> mu.X
[1] -0.73
```

## Variance

$$\sigma_X^2 = \mathrm{Var}[X] = (x_1 - \mu_X)^2 \times \Pr(X = x_1)$$
$$+ \ldots + (x_k - \mu_X)^2 \times \Pr(X = x_k)$$
$$= \sum_{i=1}^{k} (x_i - \mu_X) \Pr(X = x_i)$$

```
> var.X <- sum((X.outcomes-mu.X)^2*X.prob)
> var.X
[1] 7.9371
> sd.X <- sqrt(var.X)
> sd.X
[1] 2.817286
```